

Front

Back



Holistica JointResQ Capsule

Capsule source: Bovine

Composition:	
Each JointResQ Capsule contains:	
● Glucosamine Sulphate Sodium Chloride <i>(Derived from seafood)</i> <i>(Equivalent to Glucosamine Sulphate 250mg)</i>	314mg
● Chondroitin Sulphate Sodium <i>(Derived from chicken)</i>	200mg
● Methylsulfonylmethane (MSM)	100mg

Glucosamine is able to stimulate compounds in our bodies (glycosamino-glycans or proteologycans) that are able to restore damaged cells of our cartilage tissue (chondrocytes) while also exhibiting some anti-inflammatory abilities.

Chondroitin is a complex carbohydrate that is essential in the synthesis of cartilage-building compounds in our bodies. It is also important in the maintenance of healthy cartilage as it facilitates the absorption and retention of water in our joints.

Methylsulfonylmethane (MSM) is a naturally occurring sulphur compound commonly found in our daily diets. MSM is important in maintaining joint health and helps to form connective tissue (cartilage, tendons, and ligaments).

Indications: Adjuvant therapy for Osteoarthritis.

- Dosage:**
- Light or moderate osteoarthritis symptoms
- 2 capsules twice daily for at least 6 weeks or according to medical prescription.
- Severe osteoarthritis symptoms
- **Initial therapy:** 2 capsules thrice daily for at least 8 weeks or according to medical prescription.
 - **Follow-up therapy:** Maintenance therapy of 2 capsules twice daily should be followed for 3-4 months or according to medical prescription.

Pharmacodynamics

Glucosamine
Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysacchatides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine is naturally synthesised by the body but the ability to do so declines with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis.
Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity.

Chondroitin sulfate
The effect of chondroitin sulfate in patients with osteoarthritis is likely the result of a number of reactions including its anti-inflammatory activity, the stimulation of the synthesis of proteoglycans and hyaluronic acid, and the decrease in catabolic activity of chondrocytes inhibiting the synthesis of

proteolytic enzymes, nitric oxide, and other substances that contribute to a damaged cartilage matrix and death of articular chondrocytes. A recent review summarises data from relevant reports describing the biochemical basis of the effect of chondroitin sulfate on osteoarthritis articular tissues. The rationale behind the use of chondroitin sulfate is based on the belief that osteoarthritis is associated with a local deficiency in some natural substances, including chondroitin sulfate.

Recently, new mechanisms of action have been described for chondroitin sulfate. In an in vitro study, chondroitin sulfate reduced the IL-1-induced nuclear factor-kB (NF-B) translocation in chondrocytes. In addition, chondroitin sulfate has recently shown a positive effect on osteoarthritic structural changes occurring in the subchondral bone.

Pharmacokinetics

- Glucosamine**
- **Absorption**
After oral administration, bioavailability is low due to first-pass hepatic metabolism 26%. The gastrointestinal absorption is close to 90%.
 - **Distribution**
Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins) Volume of distribution: 2.5 Liters.
 - **Metabolism**
(Liver extensive)
The first-pass effect in the liver sees more than 70% of glucosamine metabolised.
 - **Excretion**
Renal Excretion, 10%
Feces, 11%
Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air.

Methylsulfonyl methane (MSM)
A study with Rhesus monkeys on metabolism and excretion of DMSO found the primary metabolite DMSO2 became detectable in serum approximately two hours after ingestion of DMSO. With continued DMSO ingestion, DMSO2 maintained a steady concentration in the serum. When DMSO was stopped after 14 days, the mean DMSO2 concentration declined slowly over the subsequent 96 hours, and only trace amounts were detectable after five days. The decline in serum DMSO2 was linear, and its half-life appeared to be about 38 hours. The authors observed that absorption in these animals was similar to humans, but elimination was quicker in the monkeys. DMSO2 has been shown to persist in the blood up to five times longer than DMSO.

Chondroitin Sulfate
Pharmacokinetic studies performed on humans and experimental animals after oral administration of chondroitin sulfate revealed that it can be absorbed orally. Chondroitin sulfate shows first-order kinetics up to single doses of 3,000 mg. Multiple doses of 800 mg in patients with osteoarthritis do not alter the kinetics of chondroitin sulfate. The bioavailability of chondroitin sulfate ranges from 15% to 24% of the orally administered

dose. More particularly, on the articular tissue, Ronca et al. reported that chondroitin sulfate is not rapidly absorbed in the gastro-intestinal tract and a high content of labeled chondroitin sulfate is found in the synovial fluid and cartilage.

Contraindication
It is contraindicated in patients with hypersensitivity to glucosamine sulphate, chrondroitin sulphate and methysulfonylmethane.
As the active ingredient is obtained from seafood (shark and shellfish), the product should not be given to patients who are allergic to seafood.

- Warning and precaution**
- This product treats the underlying cause of osteoarthritis and its therapeutic effect can only be seen after a few weeks. Therefore, it is advisable to take an analgesic / anti-inflammatory drug if required during the first few weeks of therapy with this product.
 - Safety and effectiveness have not been established in children (<18 years), therefore children (<18 years) should avoid using this product.
 - The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.
 - A doctor should be consulted in order to exclude the presence of other joint conditions/ diseases for which an alternative treatment should be considered.

- Interactions with other medicaments**
- ❶ Effects on glucose metabolism & antidiabetic agents: It has been hypothesised that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/ or alteration of peripheral glucose update. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness eg when used with these antidiabetic agents : Acarbose, Acetohexamide, Chlorpropamide, Glipizidede, Glyburide, Mefformin, Migitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide, Troglitazone, Glucosamine is likely safe in patients with well-controlled diabetes (HbA1 c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely / more frequently.
 - ❷ Reduced effectiveness when used with glucosamine: Doxorubicin, Etoposide, Teniposide.
 - ❸ Warfarin — Elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects — If concomitant therapy is necessary, the patient's INR should be more closely monitored.
 - ❹ Avoid combination with chitosan as it may form complexes with chondroitin sulfate, thus decreasing its absorption.

Pregnancy & Lactation
Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus.
Administration during the first 3 months of pregnancy must be avoided.

Follow-up therapy
Maintenance therapy should be followed for 3-4 months (or according to medical prescription) by administration of 2 capsules twice daily or according to medical prescription.

The treatment should be repeated every other 6 months or less according to medical prescription.

- Side Effect**
- **Cardiovascular:** Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
 - **Central Nervous System:** Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).
 - **Gastrointestinal:** Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
 - **Skin:** Skin reaction such as erythema and pruritus have been reported with therapeutic administration of glucosamine.
 - **Symptoms of over-dosage and treatment:** None known.

Route of Administration: Oral

Storage Condition: Store at room temperature below 30°C.

Shelf Life: Please refer to the packaging.

Keep out of reach of children.
Jauhkan daripada kanak-kanak.
The information contained in this leaflet is limited. For further information, please consult your pharmacist or doctor.

Packing: 30's, 60's, 100's, 120's & 180's capsules

Date of Revision: February 2024

Manufactured By:
Syarikat Wen Ken Drug Sdn. Bhd.
24, Jalan Lambak, Taman Johor,
81200 Johor Bahru, Johor, Malaysia.

Product Registration Holder:


Eldon Healthcare Sdn Bhd
200901039772 (882915-A)
No 84, Jalan I-Park I/9, Kawasan Perindustrian I-Park,
Bandar Indahpura, 81000 Kulai, Johor, Malaysia.
T 07-660 0960 E info@eldon.com.my